

Java Comparator

25 more points to get your first star!

Rank: 3047994 | Points: 0/25



Problem | Submissions | Leaderboard | Discussions | Editorial

Comparators are used to compare two objects. In this challenge, you'll create a comparator and use it to sort an array.

The `Player` class is provided for you in your editor. It has 2 fields: a `name` String and a `score` integer.

Given an array of `n` `Player` objects, write a comparator that sorts them in order of decreasing score; if 2 or more players have the same score, sort those players alphabetically by name. To do this, you must create a `Checker` class that implements the `Comparator` interface, then write an `int compare(Player a, Player b)` method implementing the `Comparator.compare(T o1, T o2)` method.

Input Format

Input from stdin is handled by the locked stub code in the `Solution` class.

The first line contains an integer, `n`, denoting the number of players.

Each of the `n` subsequent lines contains a player's `name` and `score`, respectively.

Constraints

- $0 \leq \text{score} \leq 1000$
- 2 players can have the same name.
- Player names consist of lowercase English letters.

Output Format

You are not responsible for printing any output to stdout. The locked stub code in `Solution` will create a `Checker` object, use it to sort the `Player` array, and print each sorted element.

Sample Input

```
5
amy 100
david 100
heraldo 50
aakansha 75
```

Author: Shafaet
Difficulty: Medium
Max Score: 10
Submitted By: 122340

NEED HELP?

[View discussions](#)
[View editorial](#)
[View top submissions](#)

RATE THIS CHALLENGE



MORE DETAILS

[Download problem statement](#)
[Download sample test cases](#)
[Suggest Edits](#)



```
1  import java.util.*;
2
3  class Checker implements Comparator<Player> {
4      public int compare(Player a, Player b) {
5          if (a.score != b.score) {
6              return b.score - a.score;
7          }
8              return a.name.compareTo(b.name);
9          }
10     }
11
12
13     class Player{
14         String name;
15         int score;
16
17         Player(String name, int score){
18             this.name = name;
19             this.score = score;
20         }
21     }
22
23     class Solution {
24
25         public static void main(String[] args) {
26             Scanner scan = new Scanner(System.in);
27             int n = scan.nextInt();
28
29             Player[] player = new Player[n];
30             Checker checker = new Checker();
31
32             for(int i = 0; i < n; i++){
33                 player[i] = new Player(scan.next(), scan.nextInt());
34             }
35             scan.close();
36
37             Arrays.sort(player, checker);
38             for(int i = 0; i < player.length; i++){
39                 System.out.println(player[i].name + " " + player[i].score);
40             }
41         }
42     }
43 }
```

line: 4 Col: 1

📁 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

Download

```
1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150
```

Your Output (stdout)

```
1 aleksa 150
2 amy 100
3 david 100
```